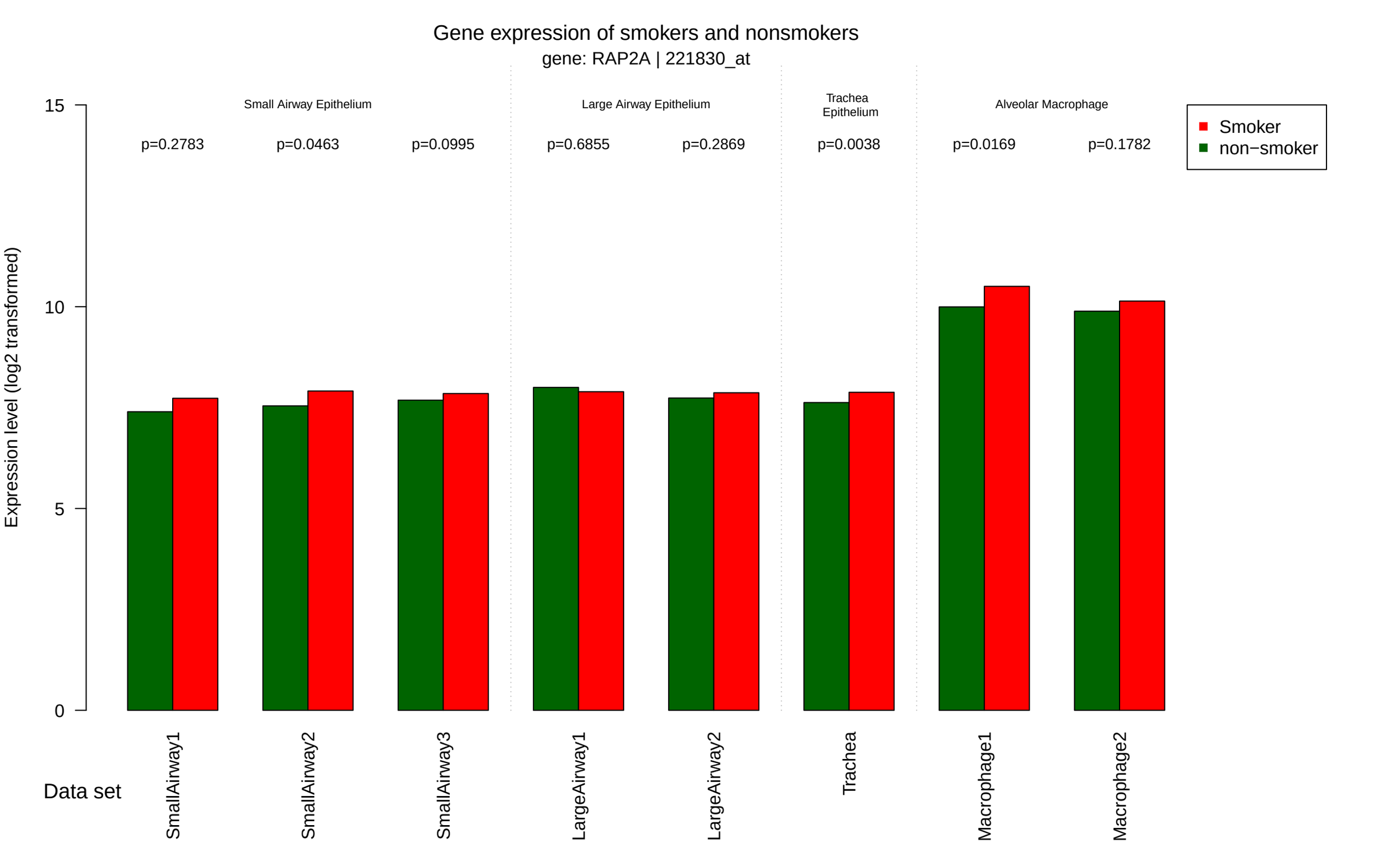
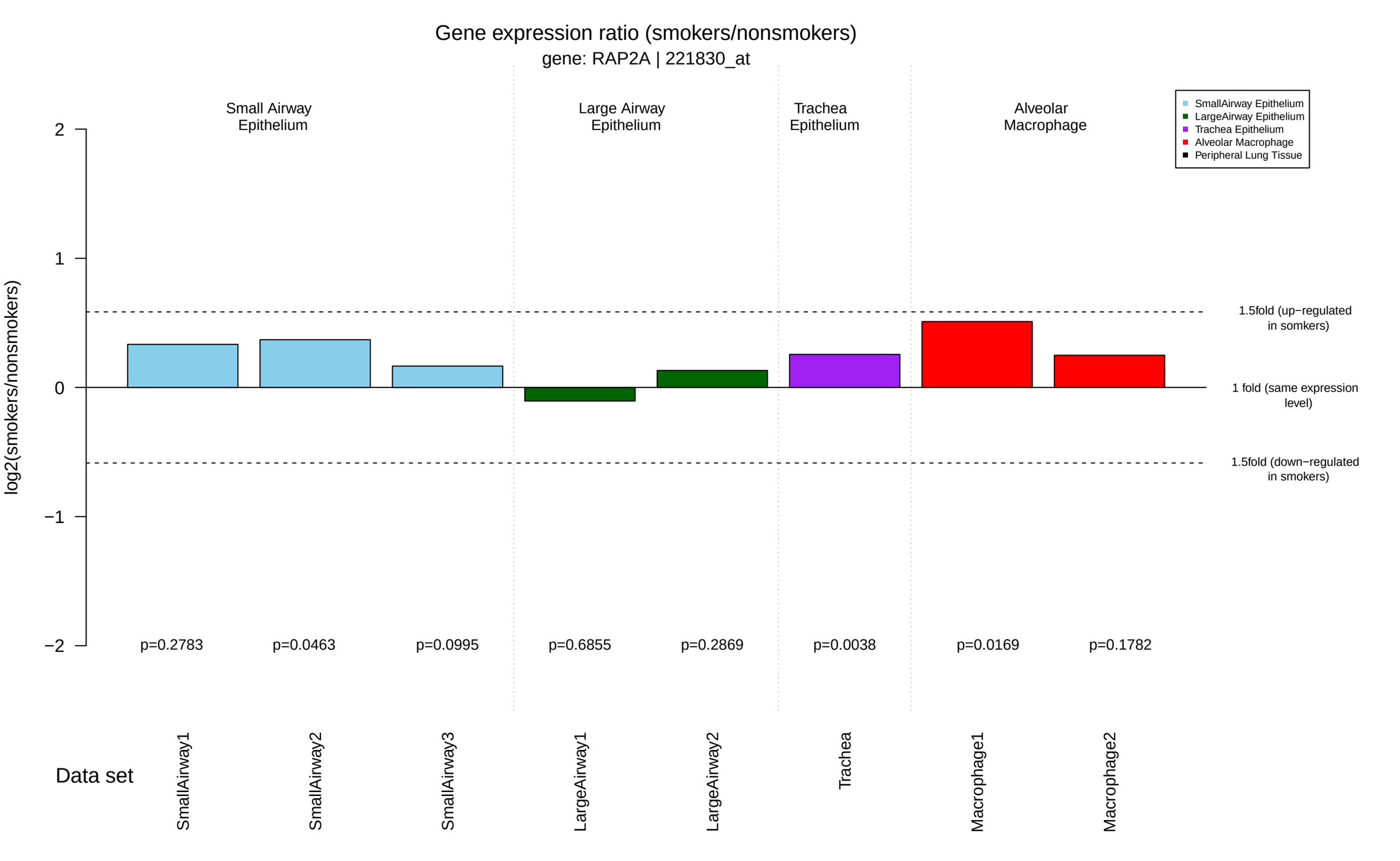
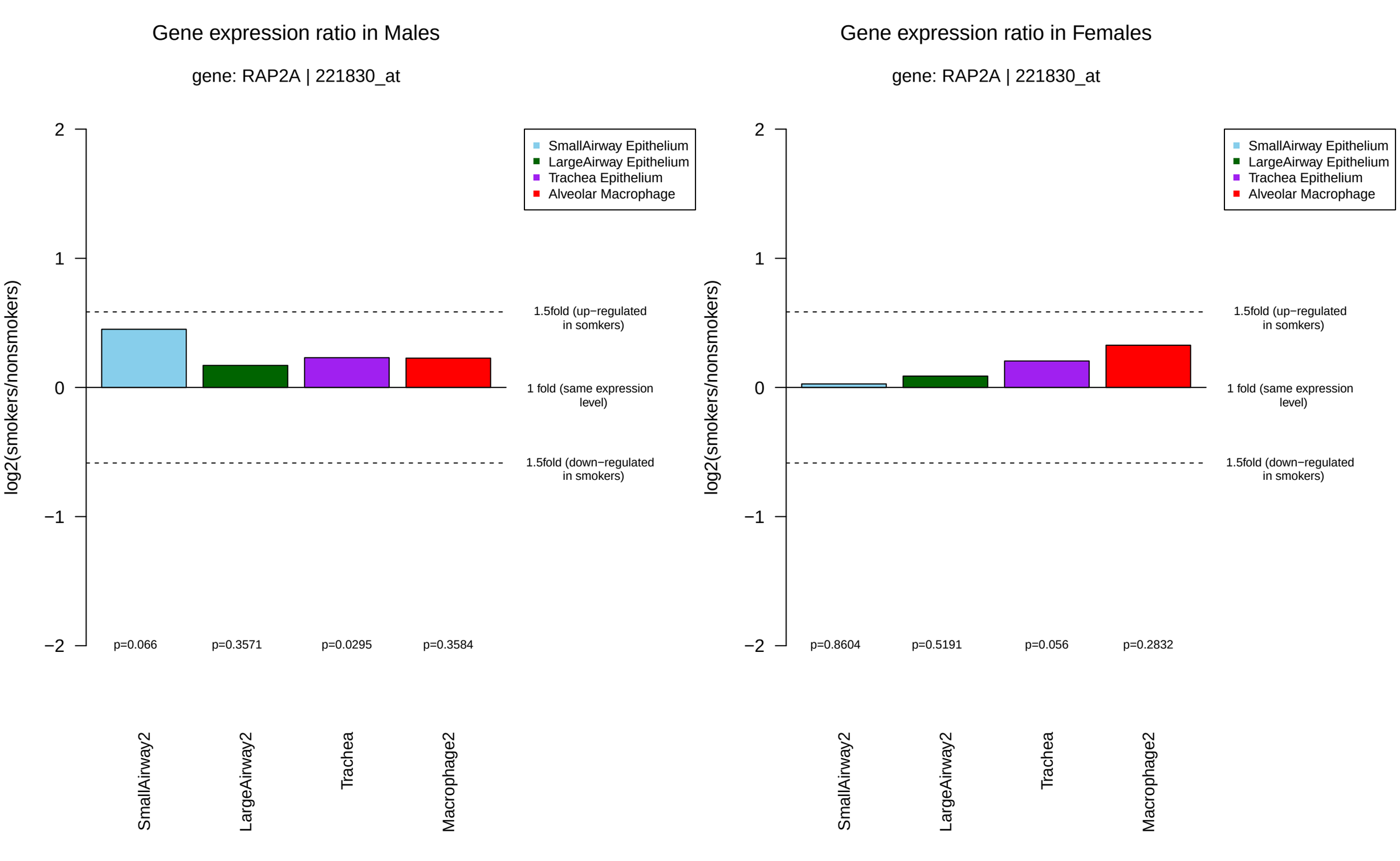
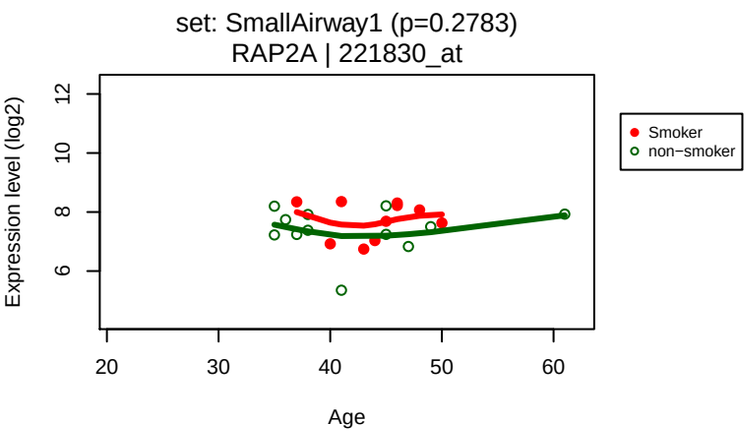
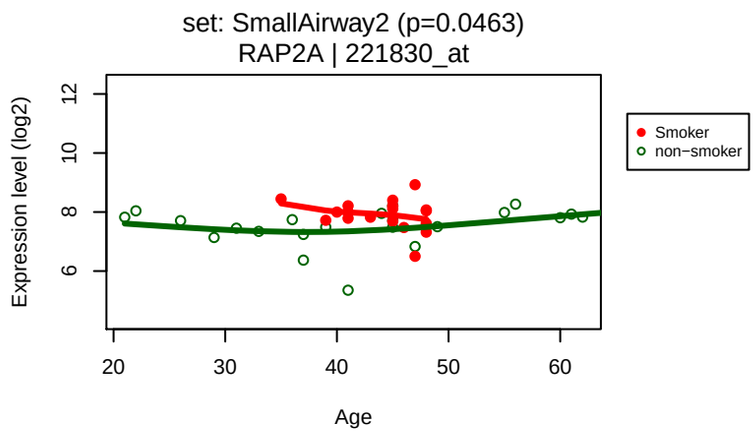
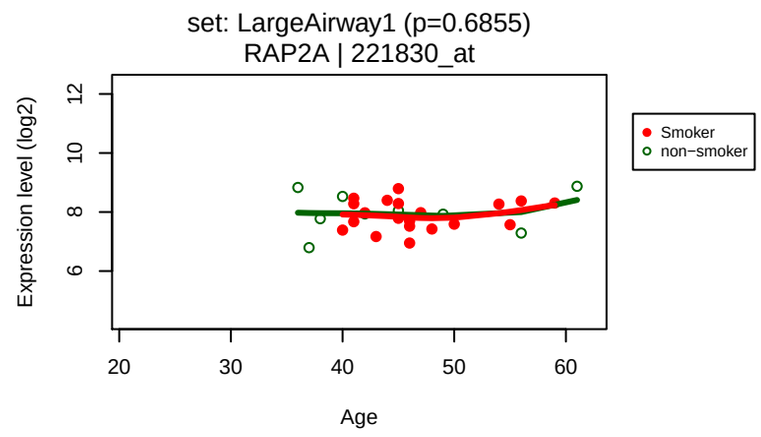

Gene expression of smokers and nonsmokers
gene: RAP2A | 221830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2783
p=0.0463
p=0.0995
p=0.6855
p=0.2869
p=0.0038
p=0.0169
p=0.1782
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

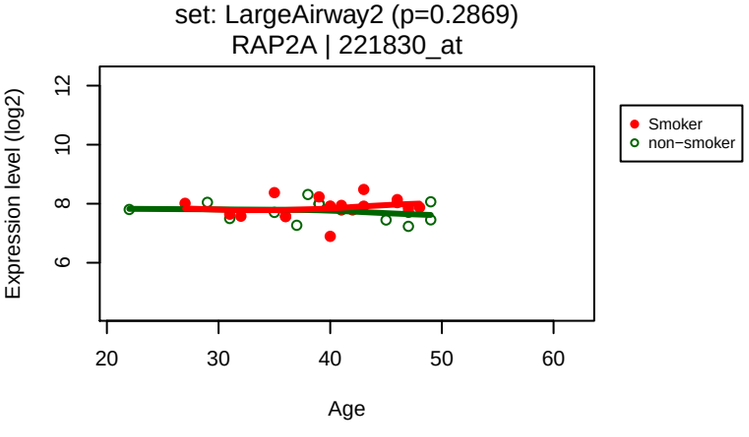

Gene expression ratio (smokers/nonsmokers)
gene: RAP2A | 221830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2783
p=0.0463
p=0.0995
p=0.6855
p=0.2869
p=0.0038
p=0.0169
p=0.1782
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

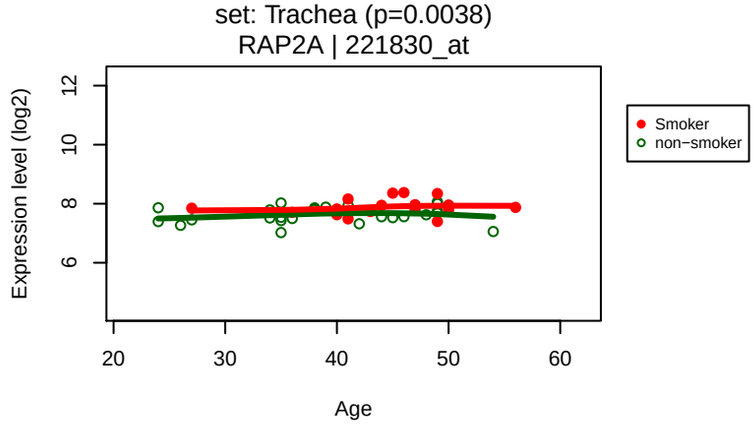

Gene expression ratio in Males
gene: RAP2A | 221830_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.066
p=0.3571
p=0.0295
p=0.3584
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAP2A | 221830_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8604
p=0.5191
p=0.056
p=0.2832
SmallAirway2
LargeAirway2
Trachea
Macrophage2

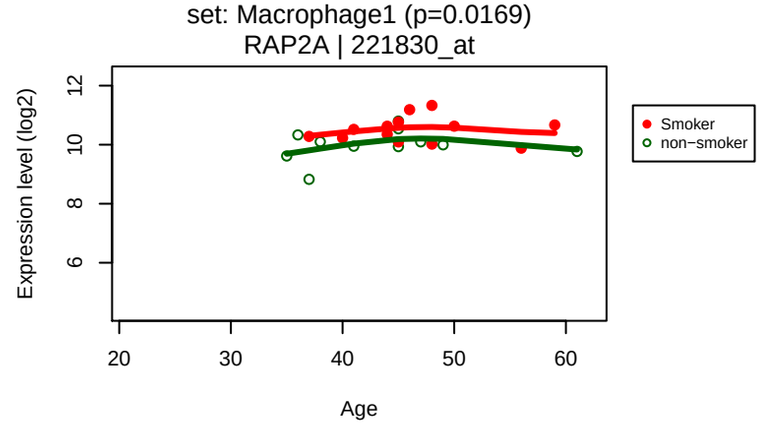

set: SmallAirway1 (p=0.2783)
RAP2A | 221830_at
Expression level (log2)
Age
Smoker
non−smoker

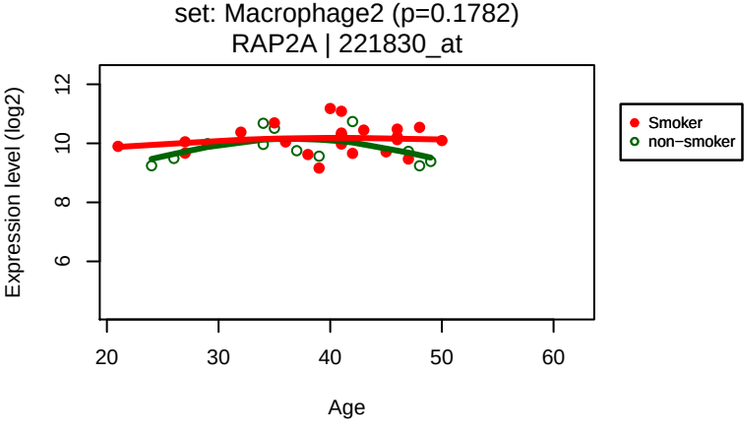

set: SmallAirway2 (p=0.0463)
RAP2A | 221830_at
Expression level (log2)
Age
Smoker
non−smoker

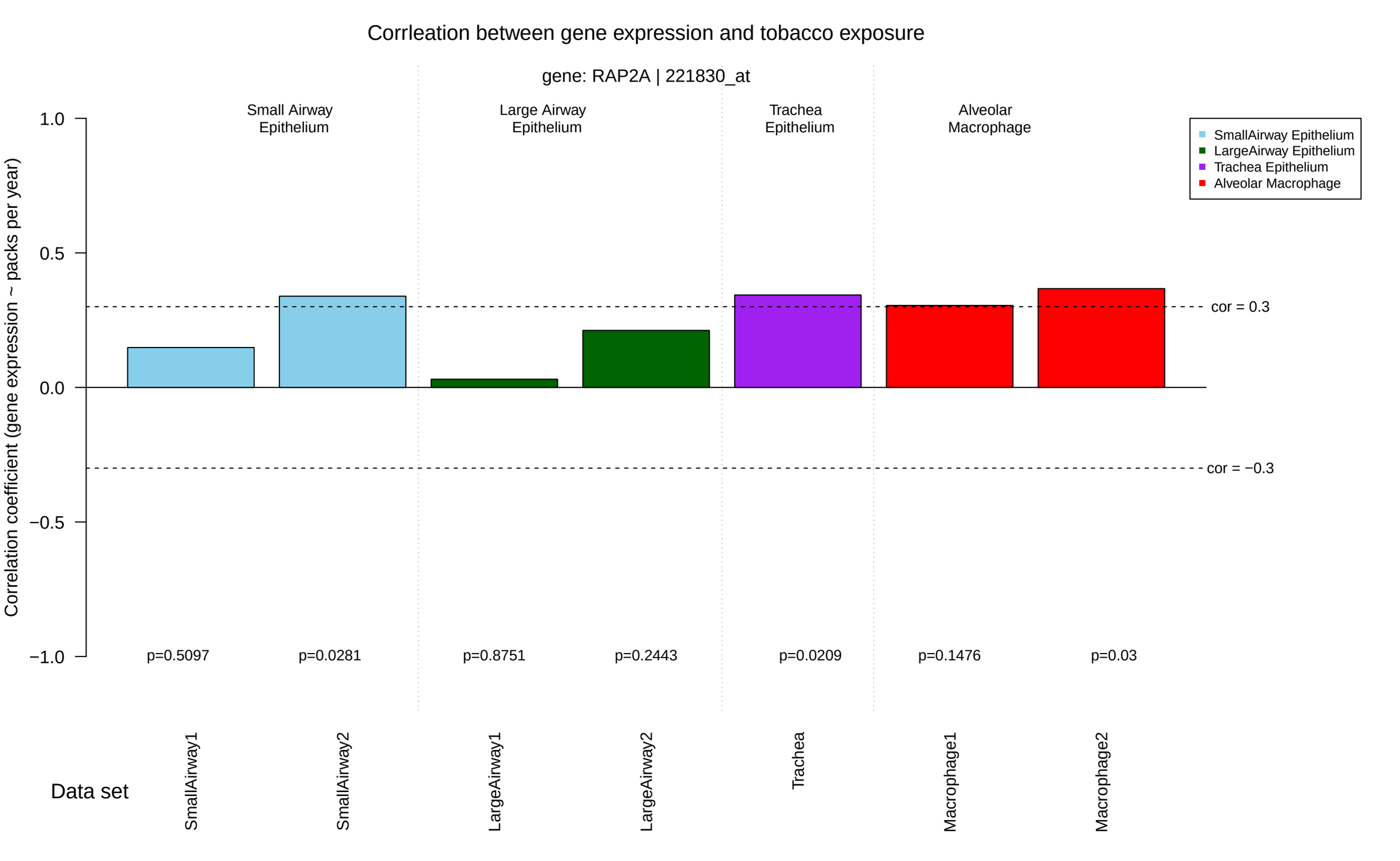

set: LargeAirway1 (p=0.6855)
RAP2A | 221830_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2869)
RAP2A | 221830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0038)
RAP2A | 221830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0169)
RAP2A | 221830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1782)
RAP2A | 221830_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAP2A | 221830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5097
p=0.0281
p=0.8751
p=0.2443
p=0.0209
p=0.1476
p=0.03
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2